JLPT Practice & Tutor

API reference | Application v0.1.0 | 18 October 2026

Implementation reference for the current FastAPI + LangChain + Ollama project. Request and response fields below are derived from the application OpenAPI schema; behavioral notes are checked against routes, validators, services and configuration.

Connection and conventions

Local base URL: http://127.0.0.1:8000. Deployment base URL: your deployed application origin. POST requests use Content-Type: application/json. Successful API responses use HTTP 200 and JSON. There is no application-level authentication declared in the current routes; OLLAMA_API_KEY is a backend-to-model proxy credential.

Interactive reference: GET /docs; alternative reference: GET /redoc; machine-readable schema: GET /openapi.json. These are FastAPI default documentation endpoints.

Method / path	Use	Model call
GET /health	Lightweight application status	No
POST /api/practice/generate	Batch vocabulary, grammar or reading questions	Yes
POST /api/practice/evaluate	Deterministic answer comparison	No
POST /api/tutor	Tutoring for a selected practice item	Yes
POST /api/practice/question	Single question, optional topic	Yes
POST /api/tutor/chat	General or question-specific tutoring	Yes
GET /	HTML practice interface	No

Integration and privacy

Use generate -> evaluate -> tutor for a complete practice flow. The bundled browser UI performs grading in browser memory and calls generate and tutor directly. The evaluate API is available to other clients. Send the answer key and explanations returned during generation when evaluating.

Study and tutor state are ephemeral: no database or persistent conversation store. Clients resend bounded tutor history with each request. Practice, tutor and root responses carry Cache-Control: no-store and Pragma: no-cache, including errors covered by middleware. Health checks do not verify model connectivity. Generated content is unofficial, difficulty is approximate, and validation does not guarantee linguistic accuracy or predict official scores.

Shared limits

Levels: N5, N4, N3, N2, N1 (case-sensitive). Explanation languages: English, Korean, Japanese; batch and tutor requests also normalize en/ko/ja and case variants. Batch categories normalize to lowercase. Batch count is 1-5, limited to 1-3 for reading; count is accepted as an alias for question_count. Tutor history permits at most 6 messages, each with role user or assistant and content 1-500 characters. Current request-body limit defaults to 4096 bytes; configuration allows 512-16384 bytes. Model timeout defaults to 45 seconds and is configurable from 1-120 seconds.

GET /health

Returns configured model name, tracing status and persistence status. base_url_configured only reports a nonempty configured URL; this is liveness/configuration information, not an Ollama readiness probe.

Response: HealthResponse

Field	Type
status	ok
model	string
base_url_configured	boolean
tracing_disabled	boolean
persistence	none

```
{
  "status": "ok",
  "model": "qwen2.5:7b",
  "base_url_configured": true,
  "tracing_disabled": true,
  "persistence": "none"
}
```

Error behavior

All POST endpoints can return 413 for oversized bodies and 422 for invalid requests. No external model call is made by this endpoint.

POST /api/practice/generate

Alias: POST /practice/generate (hidden from OpenAPI). Produces original Japanese questions with four distinct choices, one correct index and four explanations. Reading requires a Japanese passage up to 800 characters and supporting sentences in each option explanation. Stable question IDs are derived from content. Unknown request fields are rejected.

Request: PracticeGenerateRequest

See the schema catalog for every field, default and constraint. Illustrative request:

```
{
  "level": "N5",
  "category": "grammar",
  "question_count": 2,
  "explanation_language": "English"
}
```

```
curl -X POST "http://127.0.0.1:8000/api/practice/generate" \
  -H "Content-Type: application/json" \
  -d '{"level":"N5","category":"grammar","question_count":2,"explanation_language":
"English"}'
```

Response: PracticeGenerateResponse

Field	Type
level	JLPTLevel
category	GenerationCategory
question_count	integer
explanation_language	ExplanationLanguage
passage	string | null
ai_generated	true
unofficial_practice	true
difficulty_label	string
content_label	string
score_disclaimer	string
self_study_notice	string
validation	ValidationReport
questions	list[GeneratedPracticeQuestion]

Error behavior

All POST endpoints can return 413 for oversized bodies and 422 for invalid requests. Model routes also return 429 (quota/rate limit), 503 (unavailable model/service), and 502 (generation or processing failure). Timeout is explicitly mapped to 504 on batch and tutor routes; the single-question route can map timeout failures to 502.

POST /api/practice/evaluate

Alias: POST /practice/evaluate. Trims selected answer and answer key before exact comparison. options plus correct_index take precedence over correct_answer. Without a usable key, returns is_correct=false and feedback stating no answer key was provided. With four options and four explanations, returns distractor feedback. Does not call Ollama. Client-supplied keys make this self-study grading, not tamper-proof scoring.

Request: AnswerEvaluationRequest

See the schema catalog for every field, default and constraint. Illustrative request:

```
{
  "level": "N5",
  "question": "Choose the topic particle.",
  "selected_answer": "ha",
  "options": [
    "ha",
    "wo",
    "ni",
    "de"
  ],
  "correct_index": 0,
  "option_explanations": [
    "Marks the topic.",
    "Marks an object.",
    "Marks a destination.",
    "Marks a location or means."
  ]
}
```

Response: AnswerEvaluationResponse

Field	Type
is_correct	boolean
selected_answer	string
correct_answer	string
supporting_sentence	string | null
why_correct_fits	string
why_alternatives_do_not_fit	list[AlternativeExplanation]
feedback	string
grammar_notes	list[string]
self_study_mode	true
tamper_proof	false
official_score_predictive	false
difficulty_label	string
scoring_notice	string
score_disclaimer	string

Error behavior

All POST endpoints can return 413 for oversized bodies and 422 for invalid requests. No external model call is made by this endpoint.

POST /api/tutor

Alias: POST /tutor (hidden from OpenAPI). Requires selected question context (object or nonempty text up to 500 characters). question aliases: current_question, practice_item; message aliases: user_message, query; history aliases: recent_messages, messages. Unknown top-level fields are rejected. Selected context ignores extra fields and bounds option text to 200 and explanation text to 600 characters.

Request: TutorRequest

See the schema catalog for every field, default and constraint. Illustrative request:

```
{
  "level": "N5",
  "explanation_language": "English",
  "question": {
    "id": "example-question",
    "question_text": "Explain the topic particle.",
    "options": [
      "ha",
      "wo",
      "ni",
      "de"
    ],
    "correct_index": 0
  },
  "history": [],
  "message": "Why does the first option fit?"
}
```

Response: TutorResponse

Field	Type
level	JLPTLevel
explanation_language	ExplanationLanguage
question_id	string | null
reply	string
ai_generated	true

Error behavior

All POST endpoints can return 413 for oversized bodies and 422 for invalid requests. Model routes also return 429 (quota/rate limit), 503 (unavailable model/service), and 502 (generation or processing failure). Timeout is explicitly mapped to 504 on batch and tutor routes; the single-question route can map timeout failures to 502.

POST /api/practice/question

Alias: POST /practice/question. Single-question contract differs from batch generation: supports kanji as well as vocabulary, grammar and reading, and accepts an optional topic. Returns question and explanation rather than batch question_text and option_explanations. No explanation_language request field.

Request: PracticeQuestionRequest

See the schema catalog for every field, default and constraint. Illustrative request:

```
{
  "level": "N4",
  "category": "kanji",
  "topic": "daily life"
}
```

Response: PracticeQuestionResponse

Field	Type
level	JLPTLevel
category	PracticeCategory
question	string
options	list[string]
correct_index	integer
explanation	string

Error behavior

All POST endpoints can return 413 for oversized bodies and 422 for invalid requests. Model routes also return 429 (quota/rate limit), 503 (unavailable model/service), and 502 (generation or processing failure). Timeout is explicitly mapped to 504 on batch and tutor routes; the single-question route can map timeout failures to 502.

POST /api/tutor/chat

Alias: POST /tutor/chat. question is optional, allowing general tutoring. Accepts the same question aliases as /api/tutor. Its response lacks question_id and ai_generated. History is supplied by the client on every request.

Request: TutorChatRequest

See the schema catalog for every field, default and constraint. Illustrative request:

```
{
  "level": "N5",
  "explanation_language": "English",
  "message": "Explain the difference between topic and subject particles.",
  "history": []
}
```

Response: TutorChatResponse

Field	Type
level	JLPTLevel
explanation_language	ExplanationLanguage
reply	string

Error behavior

All POST endpoints can return 413 for oversized bodies and 422 for invalid requests. Model routes also return 429 (quota/rate limit), 503 (unavailable model/service), and 502 (generation or processing failure). Timeout is explicitly mapped to 504 on batch and tutor routes; the single-question route can map timeout failures to 502.

Errors and client handling

Status	Meaning / handling
200	Successful request. Inspect response fields for evaluation result.
413	Body exceeds configured byte limit. Reduce history/context or payload size.
422	Request/schema validation failed. Inspect detail entries and field locations.
429	Upstream quota or rate limit. Back off before retrying.
502	Upstream processing, structured output or validation failed.
503	Ollama service/model unavailable or configuration invalid.
504	Model request timed out on routes that map timeout explicitly.

Non-validation errors generally use {"detail": "message"}. FastAPI validation errors use a detail array. No Retry-After header is explicitly set by these routes. Retrying a generation call may produce different questions.

```
{
  "detail": [
    {
      "type": "enum",
      "loc": [
        "body",
        "level"
      ],
      "msg": "Input should be a supported JLPT level",
      "input": "N0"
    }
  ]
}
```

The 422 example above is illustrative; exact framework message wording can vary. The current Render guard rejects localhost/127.0.0.1 model URLs when RENDER=true. Use a network-reachable model endpoint for cloud deployment.

Schema catalog

The following catalog includes request models, response models and nested types exposed by OpenAPI. Required means the field must be present. Optional fields may still have defaults. Custom validation rules described earlier also apply. Descriptions and constraints follow the current source code.

AlternativeExplanation

Explanation for why a distractor option does not fit.

Field / presence	Type / constraints / default
option_index (required)	integer; minimum=0.0; maximum=3.0
option_text (required)	string
was_selected (optional)	boolean; default=false
why_it_does_not_fit (required)	string

AnswerEvaluationRequest

Request payload for deterministic, non-LLM multiple-choice answer evaluation.

Field / presence	Type / constraints / default
level (optional)	JLPTLevel; default="N5"
question (required)	string; minLength=1; maxLength=2000
selected_answer (required)	string; minLength=1; maxLength=500
correct_answer (optional)	string | null
options (optional)	list[string] | null
correct_index (optional)	integer | null
option_explanations (optional)	list[string] | null
supporting_sentence (optional)	string | null

AnswerEvaluationResponse

Local/deterministic feedback for a submitted multiple-choice answer (no LLM call).

Field / presence	Type / constraints / default
is_correct (required)	boolean
selected_answer (optional)	string; default=""
correct_answer (optional)	string; default=""
supporting_sentence (optional)	string | null
why_correct_fits (optional)	string; default=""
why_alternatives_do_not_fit (optional)	list[AlternativeExplanation]
feedback (required)	string
grammar_notes (optional)	list[string]
self_study_mode (optional)	true; default=true
tamper_proof (optional)	false; default=false
official_score_predictive (optional)	false; default=false
difficulty_label (optional)	string; default="Approximate JLPT difficulty (unofficial AI-generated estimate)"
scoring_notice (optional)	string; default="Self-study mode: The answer key is held in browser memory and compared locally without calling the Ollama model or storing scores. Because a browser-held answer key is inspectable, this scoring is for self-study feedback only, is not tamper-proof, and does not predict an official JLPT score."
score_disclaimer (optional)	string; default="Unofficial AI practice with approximate difficulty: practice accuracy is for self-study only and does not predict or guarantee an official JLPT score."

ChatMessage

Single message in an ephemeral, browser-maintained conversation window. Server-side role validation strictly permits only 'user' and 'assistant' (rejecting 'system', 'developer', or 'tool' roles to prevent prompt injection).

Field / presence	Type / constraints / default
role (required)	user / assistant; Speaker role ('user' or 'assistant' only; 'system' is forbidden).
content (required)	string; minLength=1; maxLength=500; Bounded message text content (1 to 500 characters).

ExplanationLanguage: English, Korean, Japanese

GeneratedPracticeQuestion

Validated, AI-labeled JLPT practice question returned to the client.

Field / presence	Type / constraints / default
id (required)	string; minLength=5; maxLength=80; Deterministic stable question identifier derived from content.
question_text (required)	string; Japanese question prompt text.
options (required)	list[string]; minItems=4; maxItems=4; Four distinct answer choices.
correct_index (required)	integer; minimum=0.0; maximum=3.0; Zero-based index of the single correct option (0-3).
correct_answer (required)	string; The correct option string matching options[correct_index].
option_explanations (required)	list[string]; minItems=4; maxItems=4; Explanation for each of the four options (indices 0-3).
supporting_sentence (optional)	string | null; Exact supporting sentence from the reading passage (for reading questions).
ai_generated (optional)	true; default=true; Indicates this question is unofficial AI-generated practice material.
difficulty_label (optional)	string; default="Approximate JLPT difficulty (unofficial AI-generated estimate)"; Labels difficulty as an approximate estimate.
content_label (optional)	string; default="Unofficial AI-generated original JLPT practice material (not official exam items)"; Human-readable label identifying the material as unofficial AI practice.

GenerationCategory: vocabulary, grammar, reading

HealthResponse

Health check response verifying service readiness without calling Ollama.

Field / presence	Type / constraints / default
status (optional)	ok; default="ok"
model (required)	string
base_url_configured (optional)	boolean; default=true
tracing_disabled (required)	boolean
persistence (optional)	none; default="none"

JLPTLevel: N5, N4, N3, N2, N1

PracticeCategory: vocabulary, grammar, kanji, reading

PracticeGenerateRequest

Bounded request payload for POST /api/practice/generate.

Field / presence	Type / constraints / default
level (required)	JLPTLevel; Target JLPT proficiency level (N5 through N1).
category (required)	GenerationCategory; Practice category: 'vocabulary', 'grammar', or 'reading'.
question_count (optional)	integer; minimum=1.0; maximum=5.0; default=1; Number of questions to generate (1-5 for vocabulary/grammar; 1-3 for reading).
explanation_language (optional)	ExplanationLanguage; default="English"; Language for explanations: 'English', 'Korean', or 'Japanese'.

PracticeGenerateResponse

Response schema for POST /api/practice/generate (vocabulary, grammar, and reading).

Field / presence	Type / constraints / default
level (required)	JLPTLevel
category (required)	GenerationCategory
question_count (required)	integer; minimum=1.0; maximum=5.0
explanation_language (required)	ExplanationLanguage
passage (optional)	string | null; Short original Japanese passage when category is 'reading'.
ai_generated (optional)	true; default=true
unofficial_practice (optional)	true; default=true
difficulty_label (optional)	string; default="Approximate JLPT difficulty (unofficial AI-generated estimate)"
content_label (optional)	string; default="Unofficial AI-generated original JLPT practice material (not official exam items)"
score_disclaimer (optional)	string; default="Unofficial AI practice with approximate difficulty: practice accuracy is for self-study only and does not predict or guarantee an official JLPT score."
self_study_notice (optional)	string; default="Self-study mode: The answer key is held in browser memory and compared locally without calling the Ollama model or storing scores. Because a browser-held answer key is inspectable, this scoring is for self-study feedback only, is not tamper-proof, and does not predict an official JLPT score."
validation (optional)	ValidationReport
questions (required)	list[GeneratedPracticeQuestion]; minItems=1; maxItems=5

PracticeQuestionRequest

Request payload for generating a single JLPT practice question.

Field / presence	Type / constraints / default
level (optional)	JLPTLevel; default="N5"; Target JLPT level.
category (optional)	PracticeCategory; default="grammar"; Question skill category.
topic (optional)	string | null; Optional focus topic (e.g., particles, te-form, daily life).

PracticeQuestionResponse

Structured single JLPT practice question returned by the AI service.

Field / presence	Type / constraints / default
level (required)	JLPTLevel
category (required)	PracticeCategory
question (required)	string; The JLPT multiple-choice question prompt.
options (required)	list[string]; minItems=4; maxItems=4; Four multiple-choice answer options.
correct_index (required)	integer; minimum=0.0; maximum=3.0; Zero-based index of the correct option (0-3).
explanation (required)	string; Explanation of why the correct option is right.

SelectedPracticeQuestionContext

Bounded context for the single selected practice item sent to POST /api/tutor.

Field / presence	Type / constraints / default
id (optional)	string | null
passage (optional)	string | null
question_text (required)	string; minLength=1; maxLength=500; Japanese text of the selected practice question.
options (optional)	list[string]; maxItems=4; Up to 4 options for the selected question.
correct_index (optional)	integer | null
correct_answer (optional)	string | null
selected_answer (optional)	string | null
supporting_sentence (optional)	string | null
option_explanations (optional)	list[string]; maxItems=4; Per-option explanations for the selected question.

TutorChatRequest

Stateless tutor request carrying ephemeral history from browser memory.

Field / presence	Type / constraints / default
level (optional)	JLPTLevel; default="N5"; Learner's JLPT level.
explanation_language (optional)	ExplanationLanguage; default="English"; Explanation language: English, Korean, or Japanese.
question (optional)	SelectedPracticeQuestionContext | string | null
message (required)	string; minLength=1; maxLength=500; Current learner question.
history (optional)	list[ChatMessage]; maxItems=6; Bounded recent conversation window held only in browser and request-local memory.

TutorChatResponse

Stateless tutor reply.

Field / presence	Type / constraints / default
level (required)	JLPTLevel
explanation_language (optional)	ExplanationLanguage; default="English"
reply (required)	string

TutorRequest

Bounded, stateless request for POST /api/tutor about the current practice item.

Field / presence	Type / constraints / default
level (optional)	JLPTLevel; default="N5"; Target JLPT level (N5-N1).
explanation_language (optional)	ExplanationLanguage; default="English"; Explanation language: English, Korean, or Japanese.
question (required)	SelectedPracticeQuestionContext | string; The single selected practice item (structured object or question text).
message (required)	string; minLength=1; maxLength=500; Learner's question about the selected practice item (1-500 chars).
history (optional)	list[ChatMessage]; maxItems=6; Bounded recent conversation window from browser memory (max 6 messages).

TutorResponse

Stateless tutor response for POST /api/tutor.

Field / presence	Type / constraints / default
level (required)	JLPTLevel
explanation_language (required)	ExplanationLanguage
question_id (optional)	string | null
reply (required)	string
ai_generated (optional)	true; default=true

ValidationReport

Explicit separation between structural validation and linguistic correctness.

Field / presence	Type / constraints / default
structural_validation_passed (optional)	true; default=true
linguistic_correctness_guaranteed (optional)	false; default=false
official_score_predictive (optional)	false; default=false
notice (optional)	string; default="Structural validation passed (schema, Japanese script presence, 4 distinct options, single valid answer index, and per-option explanations verified). Note: Structural validation does not guarantee linguistic nuance or pedagogical accuracy of AI-generated content."

Source and scope

Source files: app/routes.py, app/schemas.py, app/services.py, app/config.py, app/main.py and app/static/app.js. This reference documents the local checkout on 18 October 2026. Examples are illustrative and do not represent live model output. No running Ollama instance was required to build this reference.